

A disproof of Conjecture 00000004042

The values of the representation dimension are not dense above 3

1 The conjecture

Conjecture 00000004042 reads (English version):

Definition: An interval algebra is an algebra constructed from a poset with $\text{repdim} = 3$.

Conjecture: The realizable values of repdim are dense above 3, and the class of algebras with $\text{repdim} = 3$ is closed under derived equivalence.

The Chinese version (in `conjectures/00000004042.md`) says, in transliteration, “ repdim de ke shixian zhi zai 3 yishang choumi; $\text{repdim} = 3$ de daishu lei zai dao chu dengjia xia fengbi”, i.e. “the realizable values of repdim are *dense in [the region] above 3*; the class of algebras with $\text{repdim} = 3$ is closed under derived equivalence”.

The conjecture is the conjunction of two clauses:

- (1) the set V of realizable values of repdim is dense above 3;
- (2) if $\text{repdim } A = 3$ and B is derived equivalent to A , then $\text{repdim } B = 3$.

The preliminary “Definition” only introduces the name “interval algebra”. Neither clause mentions interval algebras, so the definition plays no role.

Result. Clause (1) is false: repdim takes values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ (and $-\infty$ for the zero algebra under Mathlib’s conventions), so the nonempty open interval $(13/4, 15/4) \subset (3, \infty)$ contains no realizable value. Hence the conjunction (1) $\wedge$ (2) is false, whatever clause (2) means. The proof is formalized in Lean 4 with Mathlib (Section 5).

2 Reading

repdim. “ repdim ” is Auslander’s representation dimension, the standard meaning of this abbreviation in the representation theory of algebras (Definition 2). The algebras are finite-dimensional algebras over a field, the setting of the definition quoted below from Erdmann–Holm–Iyama–Schröer.

“Dense above 3”. “Dense” is a technical term with a standard meaning. Wikipedia, *Dense set*: “a subset A of a topological space X is said to be dense in X if every point of X either belongs to A or else is arbitrarily “close” to members of A ” and “Formally, A is dense in X if the smallest closed subset of X containing A is X itself.” The Chinese text uses the construction “zai X choumi” (“dense in X ”), which names the ambient space X , here “3 yishang” (“above 3”). The realizable values are numbers, so the ambient space is the half-line of real numbers above 3. Clause (1) therefore says:

- (1) $V \cap (3, \infty)$ is dense in the topological space $(3, \infty) \subset \mathbb{R}$.

Equivalently, every nonempty open subset of $(3, \infty)$ contains a realizable value. This is exactly what the Lean definition `DenseAbove3` states, using Mathlib’s `Dense` in the subspace `Set.Ioi 3` of $\mathbb{R}$.

Two remarks on this reading.

- Density passes to open subspaces: if V were dense in $[3, \infty)$, or in the extended half-line $(3, +\infty]$, then it would be dense in the open subspace $(3, \infty)$. So the statement refuted here is implied by these variants, and the refutation covers them too.
- Scope of this reading. A different, more charitable reading is conceivable: if the ambient space were taken to be the discrete set of integers ≥ 3 , density there would amount to “every integer ≥ 3 is a realizable value”. Neither language version names such an ambient space or replaces density by integer realizability, and the realizable values are compared with the real number 3; we therefore take the standard real-topological meaning stated above as the literal one. We state explicitly that this submission does not address the integer-realizability reading (nor clause (2) on its own); it refutes the conjecture as written, under the standard meaning of “dense”. Likewise “dense above 3” is not read as “densely ordered”: the construction “dense in X ” (“zai X choumi”) is the topological notion relative to the named ambient space X .

Clause (2). Following Wikipedia, *Tilting theory*, “A and B are derived equivalent (i.e. the derived categories $\text{Db}(\text{A-mod})$ and $\text{Db}(\text{B-mod})$ are equivalent as triangulated categories)”. Because clause (1) is false, the conjunction is false whatever clause (2) says. The Lean development makes this explicit. It proves $\neg(\text{DenseAbove3} \wedge P)$ for *every* proposition P (`conjecture_00000004042_false_of_any_clause2`). The main theorem is the instance of this at a concrete formalization `ClosedUnderDerivedEquivalence` of clause (2), built on Mathlib’s derived categories (Section 5). So the refutation does not depend on any choice made in formalizing clause (2): any other formalization (bounded derived categories of finitely generated modules, k -linear equivalences, ...) is also an instance of the general theorem. The proof never uses the content of clause (2), so no fact about derived equivalence is needed.

3 Definitions and sources

All quotations below are verbatim from the retrieved sources. Mathematical symbols are transcribed into L^AT_EX.

Definition 1 (Projective and global dimension). The projective dimension $\text{pd } X$ of a module X is the minimal length of a projective resolution of X , or ∞ . Wikipedia, *Resolution (algebra)*: “The minimal length of a finite projective resolution of a module M is called its projective dimension” and “If M does not admit a finite projective resolution then the projective dimension is infinite.” The (left) global dimension of a ring R is

$$\text{gl. dim } R = \sup\{\text{pd } X : X \text{ a left } R\text{-module}\}.$$

Wikipedia, *Global dimension*: the global dimension “of a ring A denoted $\text{gl dim } A$, is a non-negative integer or infinity which is a homological invariant of the ring. It is defined to be the supremum of the set of projective dimensions of all A -modules.”

In Lean we use Mathlib’s `CategoryTheory.projectiveDimension X`, which takes values in `WithBot ℕ∞ = {⊥} ∪ {0, 1, 2, ...} ∪ {∞}`. Its docstring reads: “We also define the projective

dimension in `WithBot  $\mathbb{N}_\infty$` as `projectiveDimension`, `projectiveDimension X =  $\perp$` iff `X` is zero and behaves as expected on non-negative values”. (“`X` has projective dimension $< n$ ” means that $\text{Ext}^i(X, -) = 0$ for all $i \geq n$.) The value $\perp$ occurs only for the zero module.

Definition 2 (Generator-cogenerator, representation dimension). Let A be a finite-dimensional algebra over a field k , and let $D = \text{Hom}_k(-, k)$. For an A -module M , `add M` is the class of direct summands of finite direct sums M^n . A *generator-cogenerator* is a finitely generated A -module M with $A \in \text{add } M$ and $D(A_A) \in \text{add } M$ (equivalently $A \oplus D(A_A) \in \text{add } M$). The *representation dimension* is

$$\text{repdim } A = \inf\{\text{gl. dim } \text{End}_A(M) : M \text{ a generator-cogenerator of } A\}.$$

Sources:

- K. Erdmann, T. Holm, O. Iyama, J. Schröer, *Radical embeddings and representation dimension*, arXiv:math/0210362, Section 1 (<https://arxiv.org/html/math/0210362>): “Throughout, let k be a field, and let A be an associative finite-dimensional k -algebra. By $\text{mod}(A)$ we denote the category of finite-dimensional left A -modules. A generator-cogenerator of A is an A -module Z such that $A \oplus A^* \in \text{add}(Z)$, where $A^* = D(A_A)$ with $D = \text{Hom}_k(-, k)$ the usual duality. Recall that $\text{add}(Z)$ is the subcategory of all A -modules which are isomorphic to direct summands of finite direct sums of Z . The representation dimension of A is defined as $\text{repdim}(A) = \min\{\text{gldim}(\text{End}_A(Z)) \mid Z \text{ a generator-cogenerator of } A\}$.”
- W. Hu, C. C. Xi, *Derived equivalences and stable equivalences of Morita type, I*, arXiv:0810.4761v2, Section 5 (<https://arxiv.org/html/0810.4761v2>): “An A -module M is called a generator-cogenerator for $A\text{-mod}$ if $A \oplus D(A) \in \text{add}(M)$. The representation dimension of A , denoted by $\text{rep.dim}(A)$, is defined to be $\text{rep.dim}(A) := \inf\{\text{gl.dim}(\text{End}_A(M)) \mid M \text{ is a generator-cogenerator for } A\text{-mod}\}$. This notion was introduced by Auslander in [1] to measure homologically how far an algebra is from being representation-finite”.
- C. M. Ringel, *On the representation dimension of artin algebras*, arXiv:1107.1861v1, abstract (<https://arxiv.org/html/1107.1861v1>): “The representation dimension of an artin algebra as introduced by M. Auslander in his Queen Mary Notes is the minimal possible global dimension of the endomorphism ring of a generator-cogenerator.”

EHIS write `min`, Hu–Xi write `inf`. The proof below uses only that $\text{repdim } A$ is an infimum of global dimensions, so it applies to both formulations.

Remark 3 (Conventions that do not affect the argument). (a) A finitely generated module over a finite-dimensional algebra is finite-dimensional over k , and conversely, so “finitely generated” and EHIS’s “finite-dimensional” agree. (b) $\text{End}_A(M)$ acts on M on the left, and we use its left global dimension. $\text{End}_A(M)$ is a finite-dimensional k -algebra, hence Noetherian on both sides, and for such rings left and right global dimensions coincide (Wikipedia, *Global dimension*: “if A is a Noetherian ring, both of these dimensions turn out to be equal to weak global dimension”). In any case every global dimension, left or right, lies in $\{0, 1, 2, \dots\} \cup \{\infty\}$, and that is all the proof uses. (c) Universes. In Lean, the field k , the algebra A , the modules M , and the modules over $\text{End}_A(M)$ all live in one universe u . This loses nothing. Every finitely generated A -module is isomorphic to a quotient of some A^n , which lives in universe u , so every isomorphism class of generator-cogenerators is represented. The global dimension of a ring is determined by its cyclic modules, which are quotients of the ring and so live in its universe (Wikipedia, *Global dimension*: the right global dimension “can be alternatively defined as: the supremum of the set of projective

dimensions of all cyclic right A -modules”, and “The left global dimension of A has analogous characterizations”). All Lean statements are universe polymorphic, and the theorem is proved in every universe u .

4 The proof

Lemma 4 (Integrality). *For every finite-dimensional algebra A over a field k , $\text{repdim } A \in \{-\infty\} \cup \{0, 1, 2, \dots\} \cup \{\infty\}$. In particular, every finite real value of repdim is a non-negative integer.*

Proof. Every projective dimension lies in $W := \{\perp\} \cup \{0, 1, 2, \dots\} \cup \{\infty\}$, a complete linear order. $\text{gl. dim } R$ is a supremum of projective dimensions, so $\text{gl. dim } R \in W$. $\text{repdim } A$ is an infimum of global dimensions, so $\text{repdim } A \in W$. Under the order embedding $W \hookrightarrow [-\infty, +\infty]$ ($\perp \mapsto -\infty$, $n \mapsto n$, $\infty \mapsto +\infty$), the only elements of W that map to real numbers are the natural numbers. $\square$

Lemma 5. *No integer lies in the open interval $(13/4, 15/4)$.*

Proof. If $13/4 < m < 15/4$ then $3 < m < 4$, which is impossible for an integer m . $\square$

Theorem 6. *Clause (1) is false: the set V of realizable values of repdim is not dense in $(3, \infty)$.*

Proof. $U = (13/4, 15/4)$ is an open subset of $(3, \infty)$, and it is nonempty because $7/2 \in U$. If V were dense in $(3, \infty)$, then U would contain some $r \in V$, i.e. $r = \text{repdim } A$ for some finite-dimensional algebra A . By Lemma 4, r is an integer, and by Lemma 5, $r \notin U$. This is a contradiction. $\square$

Corollary 7. *Conjecture 00000004042 is false: for every proposition P , in particular for clause (2) however it is formalized, $\neg((1) \wedge P)$.*

5 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). `Conjecture4042.lean` imports the single source file `Conjecture4042/Basic.lean` (228 lines, namespace `Conjecture4042`, one universe variable `u`), which contains the following.

Definitions.

- `globalDimension R := supX projectiveDimension X`, the supremum (Lean `iSup`) over all $X : \text{ModuleCat.}\{u\} R$, valued in `WithBot ℕ∞` (Definition 1).
- `DualRegular k A`, the type of k -linear maps $A \rightarrow k$, i.e. $D(A_A) = \text{Hom}_k(A, k)$, with the left A -module structure $(a \cdot f)(x) = f(xa)$, defined as `f.comp (LinearMap.mulRight k a)`.
- `IsSummandOfPower X M`: there are n and A -linear maps $i : X \rightarrow (\text{Fin } n \rightarrow M)$, $r : (\text{Fin } n \rightarrow M) \rightarrow X$ with $r \circ i = \text{id}$, i.e. $X \in \text{add } M$.
- `IsGeneratorCogenerator k A M`: `Module.Finite A M`, $A \in \text{add } M$ and $D(A) \in \text{add } M$.
- `repdim k A := infM globalDimension (Module.End A M)`, the infimum (Lean `iInf`) over all $M : \text{ModuleCat.}\{u\} A$ with `IsGeneratorCogenerator k A M` (Definition 2).
- `toEReal : WithBot ℕ∞ → EReal`, the order embedding $\perp \mapsto \perp$, $n \mapsto n$, $\top \mapsto \top$, used only to compare values of `repdim` with real numbers.

- `IsRealizable v`: there are $k, A : \text{Type } u$ with `[Field k] [Ring A] [Algebra k A] [FiniteDimensional k A]` and `toEReal (repdim k A) = v`.
- `DenseAbove3 := Dense {x : Set.Ioi (3:ℝ) | IsRealizable ((x:ℝ) : EReal)}`: clause (1), density in the subspace $(3, \infty)$ of $\mathbb{R}$.
- `DerivedEquivalent A B`: there is an equivalence $e : \text{DerivedCategory (ModuleCat A)} \simeq \text{DerivedCategory (ModuleCat B)}$ of Mathlib's derived categories whose functor commutes with the shift (`e.functor.CommShift ℤ`) and is triangulated (`e.functor.IsTriangulated`). Mathlib provides the derived category of all modules. We use it because it is the derived category available in Mathlib for a noncommutative ring. This choice is not load-bearing; see Section 2.
- `ClosedUnderDerivedEquivalence`: for all finite-dimensional algebras A over k and B over k' , `repdim k A = 3 → DerivedEquivalent A B → repdim k' B = 3` (clause (2)).

Theorems.

- `exists_nat_of_toEReal_eq_coe`: if `toEReal x = r` for a real r , then $x = m$ and $r = m$ for some $m : \mathbb{N}$.
- `repdim_eq_nat_of_toEReal_eq, eq_nat_of_isRealizable`: Lemma 4 (every real realizable value of `repdim` is a natural number).
- `no_nat_in_Ioo`: Lemma 5.
- `not_isRealizable_of_mem_Ioo`: no realizable value lies in $(13/4, 15/4)$.
- `not_denseAbove3 : ¬ DenseAbove3` (Theorem 6), via `dense_iff_inter_open` applied to the open set $\{x \in (3, \infty) : 13/4 < x < 15/4\}$, which contains $7/2$.
- `conjecture_00000004042_false_of_any_clause2 (P : Prop) : ¬ (DenseAbove3 ∧ P)`.
- The main theorem

`conjecture_00000004042_false : ¬ (DenseAbove3 ∧ ClosedUnderDerivedEquivalence).`

Axiom audit. There is no sorry, no `native_decide` and no added axiom. `#print axioms conjecture_00000004042_false` reports

```
'Conjecture4042.conjecture_00000004042_false' depends on axioms:
[proptext, Classical.choice, Quot.sound]
```

and the same holds for `conjecture_00000004042_false_of_any_clause2`. The file also elaborates with `-DwarningAsError=true`. To check:

```
cd lean
lake exe cache get
lake build
```